Static and kinetic friction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish adjustable static friction from sliding friction.

SUCCESS CRITERIA - I can...

1. Compare a push with the static-friction limit.
2. Find actual friction for rest or sliding.
3. Explain friction direction using relative slipping.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A runner pushes the ground backward with **50 N**.
1. **Name the partner force**, its size and the object it acts on.
2. **Explain its role** in the runner's motion.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: 0 $\le f_s \le \mu_s N$; $f_k = \mu_k N$.
s = static; k = kinetic; N = normal force. Static friction adjusts up to its maximum. Kinetic friction applies during sliding.

Worked reasoning: A box remains still while pushed with **3 N**. Static friction is **3 N** opposite the push, provided its maximum is at least **3 N**. Static friction need not equal its maximum.

Your turn: Draw the horizontal forces. Then redraw for a **5 N** push that is still below the maximum.



Box remains at rest. Redraw for a 5 N push.

Marzano check - circle: ① ② ③

03 Below the limit

The box stays at rest with an **8 N** push right. Maximum static friction is **15 N**. **Find actual friction** and net force.



The dot represents the object. At rest; static limit 15 N

Marzano check - circle: ① ② ③

04 Crossing the limit

The push on the box in question **3** rises to **18 N**. Once it slides right, kinetic friction is **12 N**. **Complete the diagram** and **find the net force**. **Explain why you use 12 N** rather than the **15 N** static limit.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Can friction help?

A runner's foot pushes backward against the ground without slipping.
1. **Draw the friction force the** ground exerts on the foot.
2. **Explain why** "friction always points backward" is not a reliable rule.



Foot



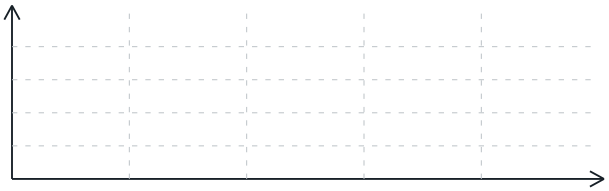
Ground

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A cart starts at **4 m** and moves right at **6 m/s** for **3 s**. **Predict final position**. Allowed endpoints are **20** to **24 m**; **state this interval** and one thing you are treating as true.
Label the diagram with the given positions and any prediction.

Position (m)



Time (s)

Marzano check - circle: ① ② ③

07 Exit check

A box has a maximum static friction of **20 N**. In one trial, a **12 N** push leaves it at rest. In a new trial, a **25 N** push starts it sliding with **16 N** kinetic friction. **Find actual friction** in the first trial and net force in the second.

Trial	Push (N)	Friction (N)	Net force (N)
At rest	12	____	____
Sliding	25	16	____

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A box has maximum static friction **8 N** and kinetic friction **6 N**.
1. **Predict the behavior** for pushes of **4**, **8** and **10 N**.
2. **Draw the sliding case**.

Add and label force arrows.

Draw the sliding case, with actual friction.

Marzano check - circle: ① ② ③